

PHYSICS SESSION ASSESSMENT: NUCLEAR REACTIONS

STUDENT NAME: _____ DATE: _____
DURATION: 1 Hour 30 Minutes SCORE: / _____

Instructions: Answer all questions clearly in the spaces provided. Show all working metrics where calculations are required.

Q1. Distinguish clearly between the processes of nuclear fission and nuclear fusion, giving one practical example or occurrence of each.

Q2. A radioactive isotope sample decays from an initial activity count of 1200 Bq down to 150 Bq over a timeframe spanning exactly 24 days. Determine the definitive half-life of this isotope.

Q3. State and briefly explain Einstein's mass-energy equivalence principle equation, detailing what each variable element represents structurally.
